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# Compare Projections.R
# Compares FloridaCommerce STWIND 2016–2024 projected growth rates
# against actual growth rates (OEWS 2016–2024) and ML model predicted growth rates.
# Comparison grain: occupation × NAICS sector (cell level)
# Metrics: RMSE, MAE, MAPE in percentage-point units; direction accuracy
# =====

# Section 0 – Setup
if (requireNamespace("rstudioapi", quietly = TRUE) && rstudioapi::isAvailable()) {
  setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
}
library(tidyverse)
library(readxl)

# =====
# NAICS join-key crosswalk
# =====
# STWIND NAICSLevel 4 codes are 6-digit strings; first 2 digits = standard NAICS sector.
# Predictions use naics_2digit where 31/32/33 (Manufacturing), 44/45 (Retail),
# and 48/49 (Transport) are each reported as separate codes.
# Recode split sectors to a single naics_key matching STWIND's grain:
#   32, 33 → 31 (Manufacturing)
#   45      → 44 (Retail Trade)
#   49      → 48 (Transportation and Warehousing)
#   99      → 90 (Government – STWIND codes as 900000 → naics_key 90)

recode_naics <- function(x) {
  case_when(
    x %in% c(32L, 33L) ~ 31L,
    x == 45L           ~ 44L,
    x == 49L           ~ 48L,
    x == 99L           ~ 90L,
    TRUE               ~ as.integer(x)
  )
}

# =====
# Section 1 – FL state projections (STWIND 2016–2024)
# =====
# NAICSLevel 4: 20 standard industry sectors
# 6-digit NAICSCode; first 2 digits crosswalk to standard 2-digit NAICS:
# 110000→11, 210000→21, 310000→31, 440000→44, 480000→48, 900000→90, etc.
# PercentChange: FloridaCommerce's own projected % change 2016→2024 (use directly)
# Exclude 000670 (Self Employed – not a NAICS industry sector)

fl_proj <- read_xlsx(
  "Data 2/STWIND_2016_2024_FL_projections.xlsx",
  guess_max = 10000
) %>%
  filter(
    NAICSLevel == 4,
    SOCLLevel == 4,
    NAICSCode != "000670",
    ConfidentialFlag == 0
  ) %>%
  mutate(
    occ_code = SOCCode,
    naics_key = as.integer(substr(NAICSCode, 1, 2)),
    stwind_pct_change = as.numeric(PercentChange) # % change 2016→2024 projected
  ) %>%
  filter(!is.na(stwind_pct_change)) %>%
  select(occ_code, naics_key, stwind_pct_change)

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message(sprintf("FL state projections loaded: %d occ x naics cells", nrow(fl_proj)))
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# =====  
# Section 1b – OEWS 2016 base employment (common anchor for growth rates)  
# =====  
# STWIND base year is 2016; use OEWS 2016 as the base for actual and model  
# growth rates so all three sources measure change over the same interval.  
  
oews_2016 <- read_xlsx(  
  "Data 2/oes_research_2016_allsectors.xlsx",  
  guess_max = 10000  
) %>%  
  mutate(across(everything(), as.character)) %>%  
  rename_with(tolower) %>%  
  rename_with(~ gsub(" ", "_", .x)) %>% # "occ code" → "occ_code", etc.  
  rename_with(~ "o_group", any_of("group")) %>% # 2016 uses "group"; 2024 uses "o_group"  
  filter(  
    area_title == "Florida",  
    o_group == "detailed"  
  ) %>%  
  mutate(  
    tot_emp_num = suppressWarnings(as.numeric(tot_emp)),  
    naics_2digit = case_when(  
      is.na(naics) | naics == "000000" ~ NA_character_,  
      TRUE ~ substr(naics, 1, 2)  
    )  
  ) %>%  
  filter(!is.na(naics_2digit)) %>%  
  mutate(naics_key = recode_naics(as.integer(naics_2digit))) %>%  
  group_by(occ_code, naics_key) %>%  
  summarise(base_2016_emp = sum(tot_emp_num, na.rm = TRUE), .groups = "drop") %>%  
  filter(base_2016_emp > 0)
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message(sprintf("OEWS 2016 base employment: %d occ x naics cells", nrow(oews_2016)))
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# =====  
# Section 2 – BLS 2024 FL actuals (sector × occupation)  
# =====  
# Use individual detailed NAICS rows (same as model training data in  
API_Data_Pull_Copy.R).  
# Sum sub-NAICS rows to naics_key grain after recoding (31/32/33→31, 44/45→44, 48/49→48).  
  
bls_2024 <- read_xlsx(  
  "Data 2/oes_research_2024_allsectors.xlsx",  
  guess_max = 10000  
) %>%  
  mutate(across(everything(), as.character)) %>%  
  rename_with(tolower) %>%  
  filter(  
    area_title == "Florida",  
    o_group == "detailed"  
  ) %>%  
  mutate(  
    tot_emp_num = suppressWarnings(as.numeric(tot_emp)),  
    naics_2digit = case_when(  
      is.na(naics) | naics == "000000" ~ NA_character_,  
      TRUE ~ substr(naics, 1, 2)  
    )  
  ) %>%  
  filter(!is.na(naics_2digit)) %>%  
  mutate(naics_key = recode_naics(as.integer(naics_2digit))) %>%  
  group_by(occ_code, naics_key) %>%
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summarise(actual_2024_emp = sum(tot_emp_num, na.rm = TRUE), .groups = "drop") %>%
filter(actual_2024_emp > 0)

message(sprintf("BLS 2024 FL actuals: %d occ x naics cells", nrow(bls_2024)))

# =====
# Section 3 – Model predictions (occ x naics sector grain)
# =====
# 1. Deduplicate any duplicate (model, occ, naics_2digit) rows by taking mean .pred
# 2. Recode naics_2digit to naics_key to match STWIND/OEWS grain
# 3. Sum recoded sub-sectors (e.g., 31+32+33 → naics_key 31)

model_preds <- read_csv("data/predictions_2024.csv", show_col_types = FALSE) %>%
  group_by(model, occ_code, naics_2digit) %>%
  summarise(pred_cell = mean(.pred, na.rm = TRUE), .groups = "drop") %>%
  mutate(naics_key = recode_naics(as.integer(naics_2digit))) %>%
  group_by(model, occ_code, naics_key) %>%
  summarise(model_pred_cell = sum(pred_cell, na.rm = TRUE), .groups = "drop")

message(sprintf("Model predictions: %d occ x naics x model rows", nrow(model_preds)))

# =====
# Section 4 – Join all sources; compute growth rates and errors
# =====
# All four sources joined on (occ_code, naics_key).
# Growth rates computed on the common 2016→2024 interval.
# Errors in percentage points (pp): predicted growth rate minus actual growth rate.

comparison <- fl_proj %>%
  inner_join(oews_2016, by = c("occ_code", "naics_key")) %>%
  inner_join(bls_2024, by = c("occ_code", "naics_key")) %>%
  inner_join(model_preds, by = c("occ_code", "naics_key")) %>%
  filter(actual_2024_emp > 0, base_2016_emp > 0) %>%
  mutate(
    actual_pct_change = (actual_2024_emp - base_2016_emp) / base_2016_emp * 100,
    model_pct_change = (model_pred_cell - base_2016_emp) / base_2016_emp * 100,
    stwind_err_pp = stwind_pct_change - actual_pct_change,
    model_err_pp = model_pct_change - actual_pct_change
  )

message(sprintf("Comparison rows: %d (occ x naics x model cells)", nrow(comparison)))

# =====
# Section 5 – Summary metrics (percentage-point units)
# =====
# RMSE_pp / MAE_pp: root mean squared / mean absolute error in percentage points
# MAPE: mean absolute % error of the growth-rate prediction (na.rm handles near-zero
# actuals)
# dir_accuracy_pct: % of cells where predicted sign of growth matches actual sign

metrics_stwind <- comparison %>%
  distinct(occ_code, naics_key, .keep_all = TRUE) %>%
  summarise(
    source = "FL State Projection 2016–2024 (FloridaCommerce STWIND)",
    n = n(),
    RMSE_pp = sqrt(mean(stwind_err_pp^2)),
    MAE_pp = mean(abs(stwind_err_pp)),
    MAPE = mean(
      pmin(abs(stwind_err_pp) / abs(actual_pct_change) * 100, 500),
      na.rm = TRUE # cap at 500pp to limit near-zero denominator
    )
  ),

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    dir_accuracy_pct = mean(sign(stwind_pct_change) == sign(actual_pct_change)) * 100
  )

metrics_models <- comparison %>%
  group_by(model) %>%
  summarise(
    source          = paste0("Model: ", first(model)),
    n               = n(),
    RMSE_pp         = sqrt(mean(model_err_pp^2)),
    MAE_pp          = mean(abs(model_err_pp)),
    MAPE            = mean(
      pmin(abs(model_err_pp) / abs(actual_pct_change) * 100, 500),
      na.rm = TRUE
    ),
    dir_accuracy_pct = mean(sign(model_pct_change) == sign(actual_pct_change)) * 100,
    .groups         = "drop"
  ) %>%
  select(-model)

metrics_all <- bind_rows(metrics_stwind, metrics_models) %>% arrange(RMSE_pp)

cat("\n--- Growth Rate Accuracy vs BLS OEWS 2024 Actuals (FL Occ x NAICS, 2016→2024) ---\n")
cat("Base year: 2016 (STWIND base / OEWS 2016); target: 2024\n")
cat("RMSE_pp / MAE_pp: error in percentage points | MAPE capped at 500pp per cell\n")
cat("dir_accuracy_pct: % of cells where projected growth direction matches actual\n\n")
print(metrics_all)

# =====
# Section 6 – Export
# =====

write_csv(comparison, "data/benchmark_comparison.csv")
write_csv(metrics_all, "data/benchmark_metrics.csv")
message("Written: data/benchmark_comparison.csv, data/benchmark_metrics.csv")

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